

CNN-Based Weed Identification for Supporting Precision Weed Control

Project Phase II Report

submitted to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the degree of

Bachelor of Technology

by

AGASTHYA HARICHANDRAN (VML22CSD008)

DIYA RADHAKRISHNAN (VML22CSD026)

MAHALAKSHMI S (VML22CSD034)

MUHASINA MUNEER V C (VML22CSD039)

under the supervision of

Ms. KARTHIKA K

Assistant Professor



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(COMPUTER SCIENCE AND DESIGN)**

VIMAL JYOTHI ENGINEERING COLLEGE (AUTONOMOUS)

CHEMPERI P.O. - 670632, KANNUR, KERALA, INDIA

MARCH 2026



VIMAL JYOTHI
ENGINEERING COLLEGE (AUTONOMOUS)
Jyothi Nagar, Chemperi - 670632, Kannur D.T, Kerala



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(COMPUTER SCIENCE AND DESIGN)

CERTIFICATE

This is to certify that the report entitled **"WEEDOUT: INTELLIGENT CROP AND WEED IDENTIFICATION WITH WEEDICIDE RECOMMENDATIONS"** submitted by **AGASTHYA HARICHANDRAN (VML22CSD008), DIYA RADHAKRISHNAN (VML22CSD026) MAHALAKSHMI S (VML22CSD034) & MUHASINA MUNEEER V C (VML22CSD039)**, to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Design is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Ms. KARTHIKA K
(Project Guide)
Assistant Professor
Computer Science and Design
Vimal Jyothi Engineering College
Chemperi

Dr. REMYA CHANDRAN
(Project Coordinator)
Associate Professor
Computer Science and Design
Vimal Jyothi Engineering College
Chemperi

Place : VJEC Chemperi
Date : 19-03-2026

Ms. NEENA V V
Program Coordinator
Computer Science and Design

(Office Seal)

DECLARATION

We hereby declare that the project report **CNN-Based Weed Identification for Supporting Precision Weed Control**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Ms. KARTHIKA K**

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma or similar title of any other University.

CHEMPERI
19-3-2026

AGASTHYA HARICHANDRAN
DIYA RADHAKRISHNAN
MAHALAKSHMI S
MUHASINA MUNEER V C

ACKNOWLEDGEMENT

Any mission never concludes without cordial co-operation from surroundings. We take this opportunity to acknowledge all the people who have helped us kind heartedly in every stages of this work. It is a matter of great pleasure for us to express our sincere gratitude to the Principal, **Dr. BENNY JOSEPH**, **Ms. NEENA V V**, Program Co-ordinator of Computer Science and Design, **Dr. REMYA CHANDRAN**, Associate Professor, Computer Science and Design, our coordinator. We are really thankful to **Ms. KARTHIKA K**, Assistant Professors, Department of Computer Science and Design, our project guides and rest of the faculty of Computer Science and Design for their valuable guidance and sincere cooperation for success of this presentation. We are thankful to our classmates and well-wishers for sharing their knowledge and suggestions. We are also thankful to our parents and the almighty.

AGASTHYA HARICHANDRAN

DIYA RADHAKRISHNAN

MAHALAKSHMI S

MUHASINA MUNEER V C

ABSTRACT

Various weeds can inhibit crop productivity due to the competition for primary requirements such as water, light, and nutrients. Using deep learning algorithms to identify and classify both crops and weed species, the project WeedOut: Intelligent Crop and Weed Identification with Weedicide Suggestions utilizes custom trained Convolutional Neural Neural Networks (CNN) to automatically classify images of crops (e.g., rice, maize, tomatoes, ginger) as well as weed varieties (e.g., broadleaf, grasses, and sedges). The information provided by this system ensures that farmers will be able to make accurate identifications of crops and invasive weed species with weedicide recommendations based on the corresponding identification. Such benefits lead to lower manual labour, increased accuracy of identification, and therefore, more sustainable practices in agriculture. This system also generates confidence scores for each prediction, allowing the user to determine the accuracy of the results; and the simple interface will provide adequate accessibility to those farmers who may not have much technical experience. This model takes real-time inputs of images and would, therefore, be conducive for use in-field as well as suggesting improvements to overall farm management through timely decision-making. Overall, the use of weedicides and the associated environmental impacts can be reduced and demonstrate how the use of AI-based technology will help develop precision agriculture.

Contents

Abstract	iii
List of Figures	viii
Nomenclature	viii
1 Introduction	1
1.1 Overview	1
1.2 General Background	1
1.3 Problem Statement	2
1.4 Scope of the system	3
1.5 Objective	3
2 Literature Review	5
2.1 Weed Net: Deep Learning Informed Convolutional Neural Network Based Weed Detection in Soybean Crops	5
2.1.1 Brief Summary	5
2.1.2 Key Studies	5
2.1.3 Gaps	6
2.1.4 Connection to Our Project	6
2.2 Forest Pest Identification Based on a New Dataset and CNN Model with Enhancement Strategy	6
2.2.1 Brief Summary	6
2.2.2 Key Studies	7

2.2.3	Gaps	7
2.2.4	Connection to Our Project	7
2.3	A Survey of Deep Learning Techniques for Weed Detection from Images	8
2.3.1	Brief Summary	8
2.3.2	Key Studies	8
2.3.3	Gaps	8
2.3.4	Connection to Our Project	9
2.4	Performance of Deep Learning Models for Classifying and Detecting Common Weeds in Corn and Soybean Production Systems	9
2.4.1	Brief Summary	9
2.4.2	Key Studies	9
2.4.3	Gaps	10
2.4.4	Connection to Our Project	10
2.5	Weed and Crop Species Classification Using Computer Vision and Deep Learning in Greenhouse Conditions	10
2.5.1	Brief Summary	10
2.5.2	Key Studies	11
2.5.3	Gaps	11
2.5.4	Connection to Our Project	11
2.6	A review on weed detection using ground-based machine vision and image processing techniques	12
2.6.1	Brief Summary	12
2.6.2	Key Studies	12
2.6.3	Gaps	12
2.6.4	Connection to Our Project	13
2.7	Weed Detection in Soybean Crops Using Custom Lightweight Deep Learning Models	13
2.7.1	Brief Summary	13
2.7.2	Key Studies	13
2.7.3	Gaps	13

2.7.4	Connection to Our Project	14
2.8	Weed Identification Using Deep Learning and Image Processing in Vegetable Plantation	14
2.8.1	Brief Summary	14
2.8.2	Key Studies	14
2.8.3	Gaps	15
2.8.4	Connection to Our Project	15
2.9	Machine Learning and Deep Learning for Plant Disease Classification and Detection	15
2.9.1	Brief Summary	15
2.9.2	Key Studies	15
2.9.3	Gaps	16
2.9.4	Connection to Our Project	16
2.10	Improving Plant Disease Classification With Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence 16	
2.10.1	Brief Summary	16
2.10.2	Key Studies	17
2.10.3	Gaps	17
2.10.4	Connection to our Project	17
3	Requirement Specification	18
3.1	Functional Requirements	18
3.1.1	User Interface	18
3.1.2	Hardware Interface	18
3.2	Non-Functional Requirements	19
3.3	Software Requirements	19
4	Proposed System and Design	21
4.1	Proposed System	21
4.2	Feasibility Study	22

4.2.1	Technical Feasibility	22
4.2.2	Operational Feasibility	22
4.2.3	Economic Feasibility	22
4.3	Design	23
4.3.1	Architecture Diagram	23
4.3.2	Data Flow Diagram	23
4.3.3	Gantt Chart	25
4.3.4	Dataset	25
5	Implementation	26
5.1	Application Development with Flutter	26
5.2	Image Acquisition and Preprocessing	27
5.3	Diagnostic and Suggestion Interface	28
6	Result and Discussion	29
7	Conclusion	33
	References	34
	Publication	36

List of Figures

4.1	Architecture Diagram	23
4.2	Level 0	23
4.3	Level 1	24
4.4	Level 2	24
4.5	Gantt Chart	25
4.6	Sample Raw Image of Crops and Weeds	25
5.1	Comparison of processed grayscale images.	28
6.1	Login Page	29
6.3	Ipomoea detected and weedicide is recommended	31
6.4	Crop classification interface	31
6.5	Performance Metrics	32
6.6	Loss curve.	32

Chapter 1

Introduction

1.1 Overview

Our project aims to improve traditional methods of identifying crops and weeds in agricultural fields, which often require significant manual effort and time. The proposed system will produce an automated, deep learning-based method of identifying and classifying agricultural crops and weeds via an analysis of field images (using a Convolutional Neural Network). This system will ultimately allow for automatic detection and classification of various crop and weed species and will significantly reduce the manual work effort that it takes for farmers to obtain accurate information on how to manage their crops more effectively and improve overall productivity in agriculture.

1.2 General Background

Weeds are a major problem in agriculture, which are the main cause of lower productivity and quality of the plants. They draw away the crops essential resources such as sunlight, water, and nutrients. Thus, the total yield may be diminished even by half.

In the past, farmers have been depending on the manual method of detecting and

removing weeds. This process is time-consuming, requires a lot of labor, and the results are usually not satisfactory, especially when dealing with enormous farm areas.

However, with the development of technology and the introduction of artificial intelligence, contemporary farming has begun the use of automated weed detection systems. The systems apply image processing and deep learning techniques to distinguish and classify weeds from crops with a high level of precision.

One of the AI techniques that have been emphasized in the last few years is Convolutional Neural Networks (CNNs) which are leading in various image classification benchmarks owing to their nature of automatic feature learning and extraction from agricultural images. Implementing such tech into precision agriculture will not only make farming efficient but also enable site-specific weed management, lower chemical use, and elevate crop yield and sustainability.

1.3 Problem Statement

In the fields of agriculture, the presence of weeds alongside crops affects the quantity and quality of the crop yield. This happens because the weeds and crops share the same nutrients, sunlight, and water. Manual weed detection, as well as removal, are slow processes that require a lot of work and are usually inaccurate, especially in big farms. Even if there are chemical ways for the control of weeds, these methods can be detrimental to the environment and the soil if the chemicals are not applied in the right amount. Hence, a need exists for a system which is an automated, accurate and efficient, and can detect weeds that lie in the same area as crops in real-time thereby distinguishing them.

The problem that is taken up for discussion in the paper is that of coming up with a model based on deep learning, using a specially designed Convolutional Neural Network (CNN) to be able to identify and classify weeds and crops by looking at the images of an agricultural setting. Employing this method is aimed at lessening the labor requirements that a farmer will have, advancing the accuracy of weed management, and along with that, raising the productivity of crops in general.

1.4 Scope of the system

The functions of the system revolve around the creation of an automated model to classify weeds and crops by applying deep learning techniques, especially a tailored convolutional neural network (CNN). It is conceived to use images from the fields, thereby accurately distinguishing the crops from the different types of weeds (like in this research the soybean field). As a result, the farmers will be empowered with a better management plan and be such in a position to act fast when the weed starts spreading.

This technology can be made to work with other plants provided that it is retrained with datasets of those crops so that it becomes compatible with different farms. Furthermore, the system also has a possible cooperative scenario with a mobile or drone-based platform where the combination of technologies allows for the on-the-spot investigation of the field situation. Additionally, it may act as a stepping stone towards the induction of a robotic system for the eradication of weeds. Finally, the system lessens the load of manual labor, improves the productivity of the land, lowers the use of harmful chemicals, and supports the adoption of environmentally safe agricultural practices.

1.5 Objective

- The first goal is to create a deep learning-based automated system that can not only identify the weeds but also classify the crop accurately. The system should be able to work with increased accuracy and efficiency.
- Secondly, a specially tailored Convolutional Neural Network (CNN) was to be realized and implemented so that it could recognize the differences among crops and various types of weeds like broadleaf and grasses.
- Besides, reducing the requirement of manual labor and hence cutting down the time of farmers used in identification and removal of weeds was also on the

agenda.

- The fourth thing is that crop yield and productivity should be improved by the use of weed management that is both timely and accurate. Apart from that, image datasets (for example, images of soybeans, soil, grass, and broadleaf) were to be used for the purpose of training and testing the model efficiently.
- Besides, the performance of the model in terms of classification accuracy and loss rate is to be considered.
- Moreover, this should open the door for the instant use of such applications with the help of a mobile device or a drone which is aiding in the task of weed detection in the field.
- The technology entails the ninth purpose of the study, i.e., the promotion of precision agriculture through the infusion of AI and image processing methods in the field of agriculture.
- Ultimately, the accurate identification of weed areas can lead to a significant drop in herbicide use, which in turn is a step towards more sustainable agriculture.
- This project will allow for the fast uploading and processing of images of fields to allow the farmer to make quick decisions for controlling weeds through timely action.

Chapter 2

Literature Review

2.1 Weed Net: Deep Learning Informed Convolutional Neural Network Based Weed Detection in Soybean Crops

2.1.1 Brief Summary

This paper illustrates how a convolutional neural network weed-detection system (CNN-WDS) can be used with UAV (Unmanned Aerial Vehicle) images of soybean (Glycine max) crops. A total of five deep learning classification models were used to evaluate their performance, including VGG16, and ResNet. The overall accuracy of these models was classified as high, with the most accurate achieving a total of 98.63%. Altogether, this research demonstrates the need for automated photography technologies in precision farming as they reduce the amount of labor required while increasing productivity. [1]

2.1.2 Key Studies

- This research looks at using CNN models like VGG16 and ResNet to classify weeds from UAV images.

- The study compares various architectures in order to find the best performing type.
- The study has a high level of precision when detecting weeds automatically.

2.1.3 Gaps

- The majority of testing has been completed with soybean crops, so it isn't clear if this model will generalize to other crops.
- High computational complexity of the model affects how useful the model will be when deployed on edge devices.
- This study does not conduct a deep analysis of how environmental variables such as light and rainy/cold weather may affect the accuracy of weed identification systems.

2.1.4 Connection to Our Project

- Potentially supports finding the most appropriate CNN architectures for weed classification.
- Aids in advancing the use of automated image-based weed identification systems.
- Suggests that balancing accuracy against computational efficiency must be done.

2.2 Forest Pest Identification Based on a New Dataset and CNN Model with Enhancement Strategy

2.2.1 Brief Summary

This document introduces a novel dataset named IP67, which contains 67,953 images of pests from 67 different species. The paper explains that by implementing one visual

saliency graph- based (GBVS) and deep learning models, we can achieve better precision in identifying the bugs. The best result was from the InceptionResNet-V2 model with an accuracy of 84.9%, which is approximately 7% higher than the baseline models. Moreover, the research also suggested a method of data augmentation to broaden the gap of the model's generalization. [2]

2.2.2 Key Studies

- A comprehensive dataset is introduced in this study (IP67) which allows for pest classification.
- Convolutional Neural Network architectures (CNN) are utilized with transfer learning techniques in order to attain improved levels of accuracy during model training phases.
- CNN models achieved increased levels of classification accuracy and generalization through training using the IP67 dataset.

2.2.3 Gaps

- Large, highly complex models require a significant amount of computing resources throughout the model training phase.
- The use of large and complex models limits the ability to interpret how a specific model or method made an individual decision.
- Fine grain classification between similar classes continues to be a challenge.

2.2.4 Connection to Our Project

- The research stresses the importance of having sufficient data (in terms of both size and diversity) available.

- Transfer learning can be used in training CNN architecture for greater levels of accuracy.
- The research provides insight into how to develop efficient and interpretable models.

2.3 A Survey of Deep Learning Techniques for Weed Detection from Images

2.3.1 Brief Summary

This report provides a comprehensive overview of recent developments in deep learning methodologies for weed identification from images (including data collection, data preprocessing, model architectures and evaluation metrics). This paper identifies challenges faced by researchers in developing these systems (such as environmental variability and the “green-on-green” challenge) and proposes future directions for research in this area, including possibilities involving the use of Generative Adversarial Networks (GANs) to develop real time applications. [3]

2.3.2 Key Studies

- The review includes CNNs, FCNs and hybrid deep learning.
- It provides information on the dataset preparation and preprocessing.
- It examines how accurate and F1-score are used to evaluate performance.

2.3.3 Gaps

- The need for large labelled datasets for deep learning models.
- The performance of a model will reduce based on changing conditions.
- High computation limitations affect real-time applications.

2.3.4 Connection to Our Project

- Helps with model and dataset selection.
- Identifies challenges related to lighting and occlusion.
- Proposes future improvements e.g. use of GANs and real-time systems.

2.4 Performance of Deep Learning Models for Classifying and Detecting Common Weeds in Corn and Soybean Production Systems

2.4.1 Brief Summary

This paper checks how well deep learning models work for classifying and detecting weeds in corn and soybean fields. Models like VGG16, which's a type of CNN are really good at classifying weeds with an accuracy of 98.9%. They used YOLOv3 model to detect weeds. The study shows that deep learning can help manage weeds in areas of a field. It is still hard to detect some types of weeds accurately. Deep learning models can help with weed classification and detection in corn fields and soybean fields. Weed classification and detection are important for site- weed management, in corn and soybean fields. [4]

2.4.2 Key Studies

- CNNs will be used to classify images and YOLO for detecting the weeds.
- The CNN model, specifically VGG, has the highest level of accuracy for image classification.

- The performances of the models have been looked over from different points using classification scores (accuracy and F1) and detection scores (mAP).

2.4.3 Gaps

- The dataset is limited to a certain number of types of weeds.
- There was a lack of accuracy detecting grassy weeds within the test dataset.
- The imbalance between the two types of weeds within the training and testing datasets will impact the model's ability to generalize.

2.4.4 Connection to Our Project

- This research supports an integrated approach; combining classification and detection methods.
- This research demonstrates that having a balanced dataset will improve the performance of a weed identification system.
- This research can contribute to improving the method used to locate weeds within an image.

2.5 Weed and Crop Species Classification Using Computer Vision and Deep Learning in Greenhouse Conditions

2.5.1 Brief Summary

The study is about how well machine learning (Support Vector Machines, SVMs) and deep learning (Convolutional Neural Networks, CNNs) can identify automatically the different types of weeds and crops in a greenhouse. Researchers training a VGG16-based CNN model on their 3,792 RGB image dataset achieved an outstanding

classification accuracy of 97.5%. This result exceeded that of SVMs by an order of magnitude. Moreover, as per their experiments, SVMs required complete feature extraction and preprocessing. As a result, the research highlights the potentials of CNNs in the simplified, on-demand scouting of crops and weeds in controlled environment agriculture. [5]

2.5.2 Key Studies

- Key findings from studies
- The study compares VGG16 and SVM as methods of classifying weeds. VGG16 classifies weeds with higher accuracy and does not require any manual feature extraction.
- The model produces good results when evaluated under a controlled greenhouse environment.

2.5.3 Gaps

- There is an imbalance in the species represented by weed datasets.
- The research is conducted only in controlled environments.
- Plant types exhibit variable levels of performance within the same experiment.

2.5.4 Connection to Our Project

- Supports the utilization of deep learning over traditional ML techniques.
- Eliminates manual intervention required for features extraction and curates high-quality data.
- Aids in the development of general random forest classification (RRF) models with high accuracy.

2.6 A review on weed detection using ground-based machine vision and image processing techniques

2.6.1 Brief Summary

The aim of this article is to produce a lightweight and computationally efficient deep learning algorithm (CNN) to efficiently detect weeds via image recognition on less powerful devices, such as a cell phone. This model has been created using a much simpler design which incorporates less layers and parameters than would usually be used in order to reduce the amount of time needed to perform a calculation while still giving accurate results. The intended use of this algorithm is to support farmers with a practical means to detect and manage weeds in real time without requiring expensive hardware. [6]

2.6.2 Key Studies

- This article evaluates the entire weed identification process.
- The comparison of classical machine-learning methods to deep-learning based weed identification has been made.
- An examination of identification problems such as varying light levels and objects blocking other weeds has been conducted.

2.6.3 Gaps

- No standardized datasets for benchmark comparisons.
- No detailed comparisons of method performance exists.
- No clear analysis of computational costs for method implementation is available.

2.6.4 Connection to Our Project

- Provides a comprehensive view of system architecture.
- Discusses the practical challenges of implementing the system.
- Suggests improvements such as use of multiple sensors for integration.

2.7 Weed Detection in Soybean Crops Using Custom Lightweight Deep Learning Models

2.7.1 Brief Summary

2.8 Lightweight CNN Models

This study is about making CNN models that can detect weeds in real time. The model we made is really good at this. Can do it fast and without using up too much memory. We made sure it can work well on devices like the Raspberry Pi. What is important here is that our model is both good at what it does and does not use up much resources, which is what we need for real world applications. Lightweight CNN models are what we are focusing on. [7]

2.7.2 Key Studies

- Crop detection followed by weed segmentation is used in this method.
- The requirement for a large training dataset is decreased.
- The model provides both high precision and recall.

2.7.3 Gaps

- Crop detection needs to be accurate for the method to work.

- Only specific crops can be used, such as bok choy.
- Annotating for large-scale use is time-intensive.

2.7.4 Connection to Our Project

- WeedOut uses a different approach to detect weeds.
- It reduces the need for large amounts of training data.
- It improves on the generalization of the model when detecting new types of weeds.

2.8 Weed Identification Using Deep Learning and Image Processing in Vegetable Plantation

2.8.1 Brief Summary

This study is about making CNN models that can detect weeds in real time. The model we made can detect weeds 97.7 percent of the time. It also works fast and does not use a lot of memory. We made this model to work well on devices like the Raspberry Pi. The important thing, about this work is that it shows we can make models that work well and are also efficient which's what we need for real world applications. [8]

2.8.2 Key Studies

- The research builds light weight CNNs suitable for use in edge devices.
- The current model has high accuracy with very low latency.
- It can successfully be used to identify weeds in real-time, on low power devices such as the Raspberry Pi.

2.8.3 Gaps

- The dataset is small in size making generalizing difficult.
- Class imbalance causes inaccurate predictions.
- No full evaluation of real-time performance under field conditions.

2.8.4 Connection to Our Project

- Helps with creating the ability to deploy on lower power, edge devices.
- Assists with building the performance of fast economical models.
- Allows for real time detection of weeds through automated systems.

2.9 Machine Learning and Deep Learning for Plant Disease Classification and Detection

2.9.1 Brief Summary

This paper looks at a lot of machine learning and deep learning models that can detect diseases in plants. We looked at models like YOLOv5 and ResNet50. Mobilenetv2 to see how well they work. We also looked at the data we used to test these models and how we measured how well they worked.. We looked at how efficient these models are. Our goal is to help people choose the models for use in agriculture by giving them information, about plant disease detection models. Plant disease detection is what we are trying to improve with these models. [9].

2.9.2 Key Studies

- Analysis included a number of different Machine Learning and Deep Learning models for the detection of plant cases of disease

- Rough comparisons of models such as YOLOv5 vs ResNet50 were made
- Assessment of both datasets and performance metrics were analysed in conjunction with one another.

2.9.3 Gaps

- These studies did not focus solely on weed detection
- No new model proposal was implemented
- High Complexity of Implementation were noted.

2.9.4 Connection to Our Project

- Provides a comparative baseline among various models used in the weed detection field
- Will assist with providing direction for model selection
- Will support the use of existing classification/detection models.

2.10 Improving Plant Disease Classification With Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence

2.10.1 Brief Summary

This research suggests an approach to classification of plant disease using deep learning methods and EfficientNetB0 that includes Explainable Artificial Intelligence (XAI) with the LIME framework. The model makes predictions for 38 diseases using evaluation metrics in performance; accuracy (99.69%), precision (98.27%) and recall

(98.26%). To enhance usability of the model in real-time, an app called PlantCare was developed for mobile devices which provides detection of plant disease in the field. [10]

2.10.2 Key Studies

- Utilization Of EfficientNetB(0) In Disease Classification
- Application Of The Local Interpretable Model-Agnostic Explanation (LIME) As An Explainable AI Tool
- Achievement Of High Classificatory Accuracy.

2.10.3 Gaps

- Dataset Does Not Represent The Various Environmental Factors
- Too Few Models Evaluated
- Performance Is Decreased When Evaluating In Low Lighting Situations.

2.10.4 Connection to our Project

- Implementation Of Explainable AI Improves Transparency
- Increased Level of Confidence From Users On AI Predictions
- Improved Decision Making For Weed Detection Systems.

Chapter 3

Requirement Specification

3.1 Functional Requirements

3.1.1 User Interface

Users will have a smooth interface to communicate with the WeedOut system, with overwritten prompts and instructions. Users can simply register and securely login, upload images of crops or leaves, and view results, including weed/crop identification, confidence scores, and recommended weedcides. The interface needs to be comfortable for farmers with limited technical background, meaning results must be displayed clearly and confirmation must be provided to acknowledge that images are uploaded.

3.1.2 Hardware Interface

Ask users for permission to use their devices camera to take photos. Users must have the option to edit image quality and capture options. The device must build a system that supports smartphones, tablets, and desktops. Images must be optimized for being classified with the highest quality, highest and clear to best classify.

3.2 Non-Functional Requirements

- **Performance:** The optimization of the imaging classification system will provide an efficient way of improving the CNN models' performance and database query performance. Minimizing the complexity of processing the data will minimize the amount of time it takes to upload an image and to generate a classification result from an image. Implementing caching for frequently accessed data as well as optimizing processing tasks is intended to yield faster response times for users and provide a system that works smoothly.
- **Scalability:** To ensure additional resources or servers can be added without degrading the performance of the service, the system architecture will be designed in a modular fashion. Load balancing may be implemented to help enable a more efficient distribution of work and containerization technologies may be utilized for deployment and auto-scaling based on demand and performance metrics.
- **Security:** The implementation of secure authentication and authorization mechanisms for the protection of user data is necessary in order to protect the system and the data it contains. In order to accomplish this, all communications between the systems will be encrypted and secure software development lifecycle (SDLC) guidelines will be followed during the development of the system. Periodic monitoring of the system, performing routine security audits, and conducting penetration tests will be performed to maintain system integrity and to provide protection from potential threats.

3.3 Software Requirements

- **Programming Language:** The Python was chosen due to its considerable data science ecosystem (NumPy, Pandas) and simple to understand syntax which ultimately facilitates faster prototyping of Convolutional Neural Networks (CNN) and easier integration with back end systems.

- **Web Framework:** Django is used to develop our web interface. Using a comprehensive framework allows us to have built in security and allows us to use an Object-Relational Mapping (ORM) to simplify database interactions. As a result, the data from the user upload can easily be translated into recommendations for the weedicide prediction engine.
- **Database:** MySQL is a Relational Database Management System (RDBMS). This database is used to store data related to user accounts, all the previous identification attempts, and the recommendation table for weedicide.
- **Deep Learning:** The Tensorflow and Keras Frameworks will be used to develop our CNN. They both offer the high-level programming constructs needed to tune hyperparameters and ensure the highest accuracy classification.
- **Computer Vision:** Image acquisition and pre-processing of data will occur here. We will perform a number of critical processing functions on the input images such as to correct for sizing, apply noise reduction, and convert colours to grayscale; all of which will help to standardise the input to our CNN.
- **IDE:** The primary development platform for this project will be PyCharm which was selected due to the included Python debugging tools, the included environment management tools and its deep integration with professional development projects.

Chapter 4

Proposed System and Design

4.1 Proposed System

- Use a deep learning model with the use of CNNs to classify the images to crop or weed.
- Images are preprocessed into grayscale, resizing, normalization, and augmentation to improve accuracy.
- Have a web app with authentication to register users and let a user upload an image, running on the backend side with Django.
- Provide a Weedicide Recommendation Module that will suggest potential treatments based on the weed detected.
- Store user data, images, and treatments with a MySQL database.
- Provide the end user the results, including classification, confidence score, suggested treatment, as well as maintain it with a secure access.
- Future proof the design by expanding the dataset, user feedback, while considering mobile integration that can detect the classification in real time.

4.2 Feasibility Study

4.2.1 Technical Feasibility

In principle, the WeedOut system is technically feasible because it has been created by using open-source platforms available without licensing restrictions. The model is trained and deployed with Python, Django, and TensorFlow which efficiently run on the platforms used. The InceptionV3 CNN model is a proven architecture that works well for feature extraction and classification. The backend of the platform can be hosted on any standard server, while Flutter is used for a good interface on mobile devices.

4.2.2 Operational Feasibility

WeedOut has operational feasibility characterized by an easy-to-use interface which is designed for farmers or agricultural specialists. The users can easily upload photos, see whether a weed is detected, and be provided with relevant weedicide recommendations. This helpful interface reduces the necessity for any technical knowledge, creating easy acceptance in rural and under-resourced settings. The design of this simple interface will enhance operational feasibility by producing an effective system and fit in with everyday farming activities.

4.2.3 Economic Feasibility

WeedOut is economically justified through the use of open-source tools which contribute to significant savings in development expenses. Ongoing expenses are for things like data storage fees, costs associated with retraining and re-evaluating the models used, and maintenance of the system over time. For farmers, there is real economic return for reduced weed management costs, reduced crop loss (yield), and yield efficiency. This economic argument justifies the sustainability and economic viability of WeedOut to precision agriculture.

4.3 Design

4.3.1 Architecture Diagram

An architectural diagram simplifies complex systems by highlighting key components and their interactions. It serves as a blueprint for developers and business leaders, aiding in communication and decision-making throughout the system's lifecycle.

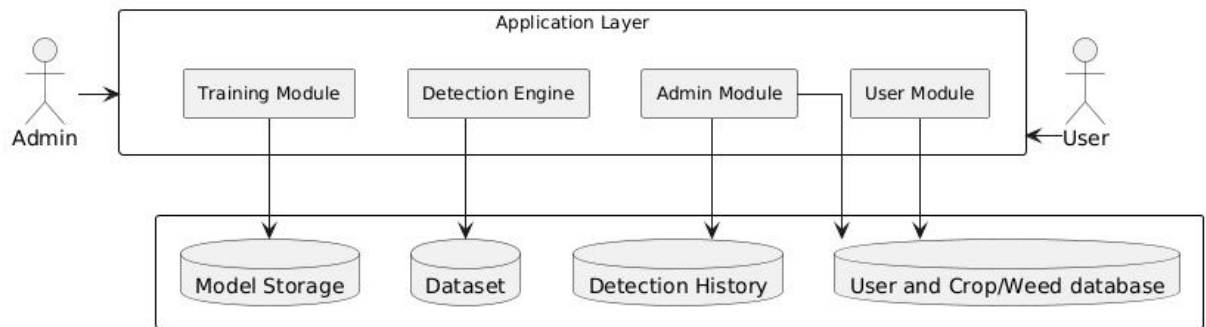


Figure 4.1: Architecture Diagram

4.3.2 Data Flow Diagram

A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. It provides information about how data enters and exits the system, changes made on the system, and where the data is stored.

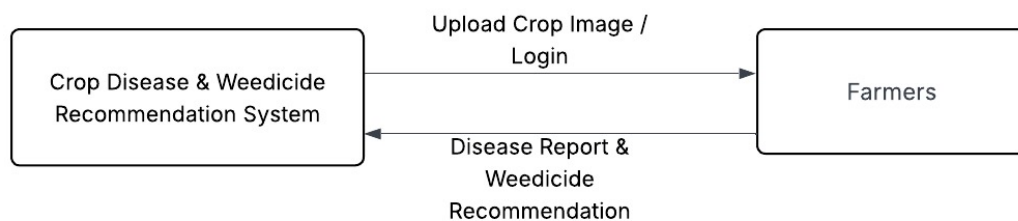


Figure 4.2: Level 0

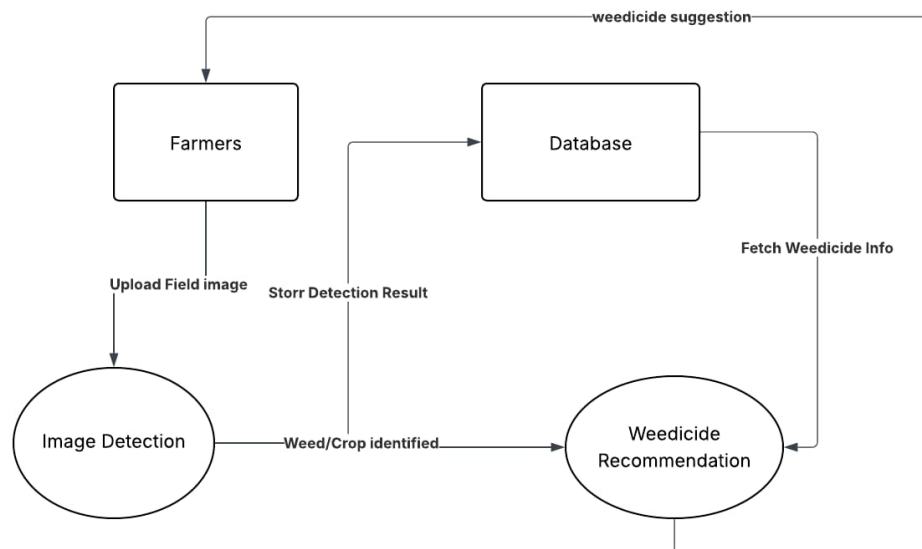


Figure 4.3: Level 1

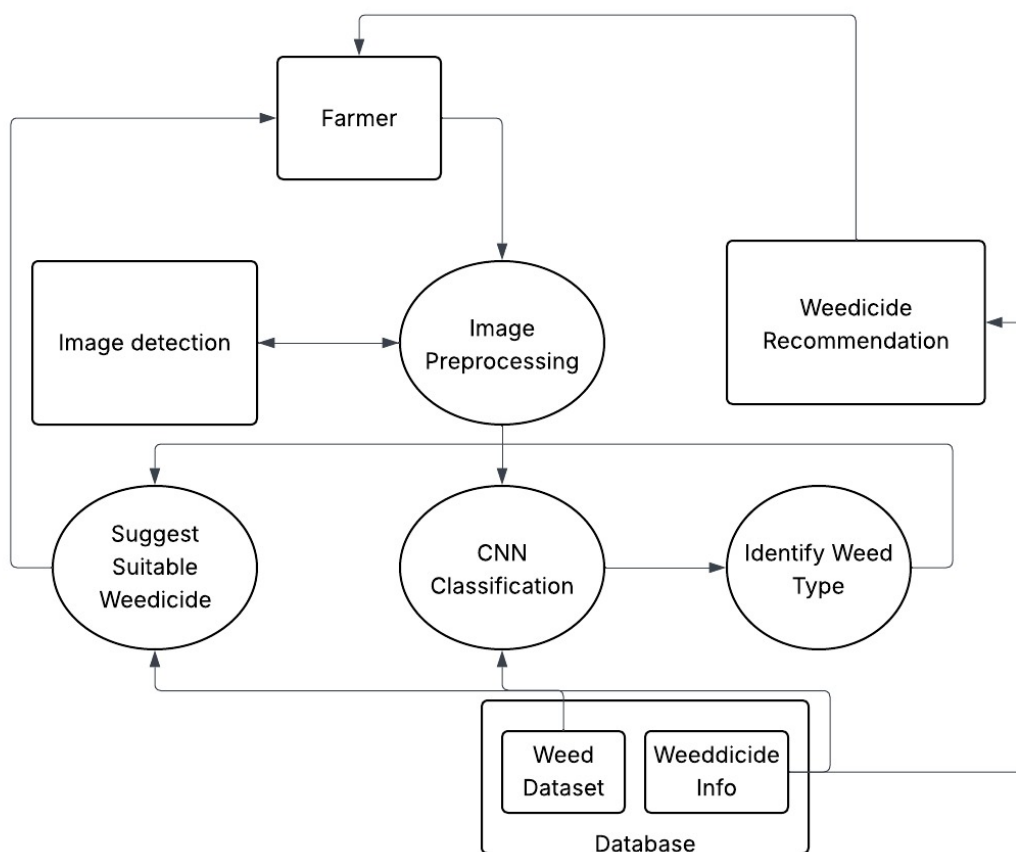


Figure 4.4: Level 2

4.3.3 Gantt Chart

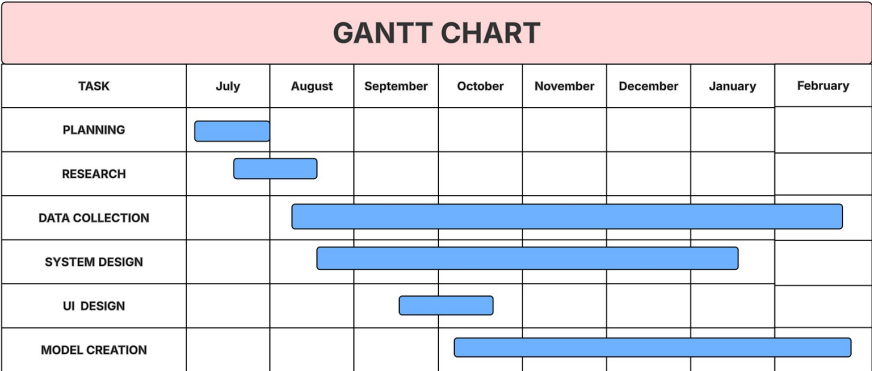


Figure 4.5: Gantt Chart

4.3.4 Dataset

The dataset represents the base level of knowledge for the system and provides the necessary morphometric attributes of organisms during various growth and development stages, as well as under different environmental conditions. It provides the main input to the CNN, therefore determining how the model will learn and classify plants correctly. This library serves as the model for the system to work from, thereby ensuring accurate identification in real-world agricultural environments.

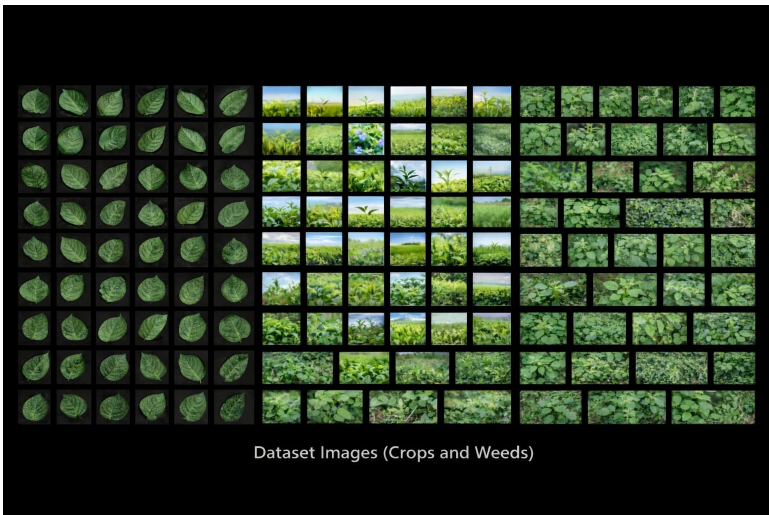


Figure 4.6: Sample Raw Image of Crops and Weeds

Chapter 5

Implementation

5.1 Application Development with Flutter

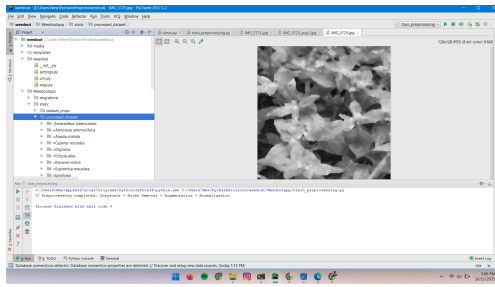
Using Flutter and Dart as the underlying programming languages, WeedOut became an application development project that could be used by both Android and iOS users with one codebase due to the cross-platform capability of the framework.

- **Cross-Platform Application Support:** With the use of Flutter, the application will have a consistent user experience across multiple devices, thus giving access to a wider range of farmers.
- **UI and UX Design:** The Flutter framework provides a large number of UI Components (Widgets), making it possible to build a responsive and user-friendly UI allowing the user to easily navigate through their application and submit images from their fields.
- **State Management:** The WeedOut app employs an effective state management approach to facilitate the real-time selection/sourcing of images, processing the selected image(s), and display of completed images back to the user while maintaining acceptable system performance throughout each of these tasks.

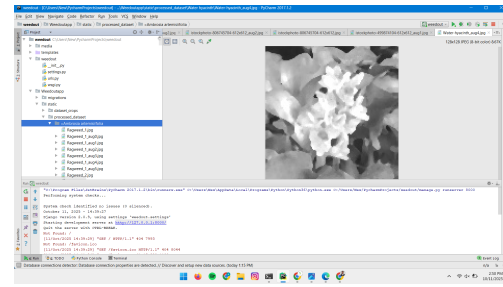
5.2 Image Acquisition and Preprocessing

A core feature of the app is the ability to acquire field images, which serve as the primary input for the identification model.

- **Camera and Gallery Access:** The app's primary input for the classification model is the field image obtained through image acquisition. The application obtains the field images using Flutter plugins, either by taking a live photograph of an object in the field or by selecting the image stored in the device's gallery.
- **Image Selection and Validation:** Images selected by the user are automatically validated in order to assure that the image has a plant in it to meet the requirements for classification purposes. Users will be notified of the selection of an image by an alert if the image selected does not have a plant in it.
- **Image Preprocessing:** The image preprocessing functions performed on the images before being processed by the CNN model are as follows:
 - **Resizing:** The images are resized to a standardized dimension that matches the input dimension requirement of the model.
 - **Normalization and Noise Removal:** The visual noise is reduced from the image to increase the accuracy of the feature extraction.
 - **Gray Conversion:** The images are converted to gray for certain stages of analysis and to enhance the visible leaf patterns and textures.



(a) Grayscale image of a weed



(b) Grayscale image of a crop

Figure 5.1: Comparison of processed grayscale images.

5.3 Diagnostic and Suggestion Interface

As of now, you can also enter an image of a diagnostic to get a detailed identification of the crop or weed available to you.

- **Results of Identification:** Once you receive a diagnostic you will receive the plant name (for example, "Ipomoea" or "Garlic") with a confidence number (percent chance).
- **Weedicide Suggestions:** You will also receive weedicide recommendations based on detected species of weeds to help you control your crop immediately.
- **Navigation:** You may also use the system to either review your previous plant identification history or to start a new plant identification task.

Chapter 6

Result and Discussion

Screenshots of the system's user interface have been used to illustrate the output that was obtained after providing the necessary input.

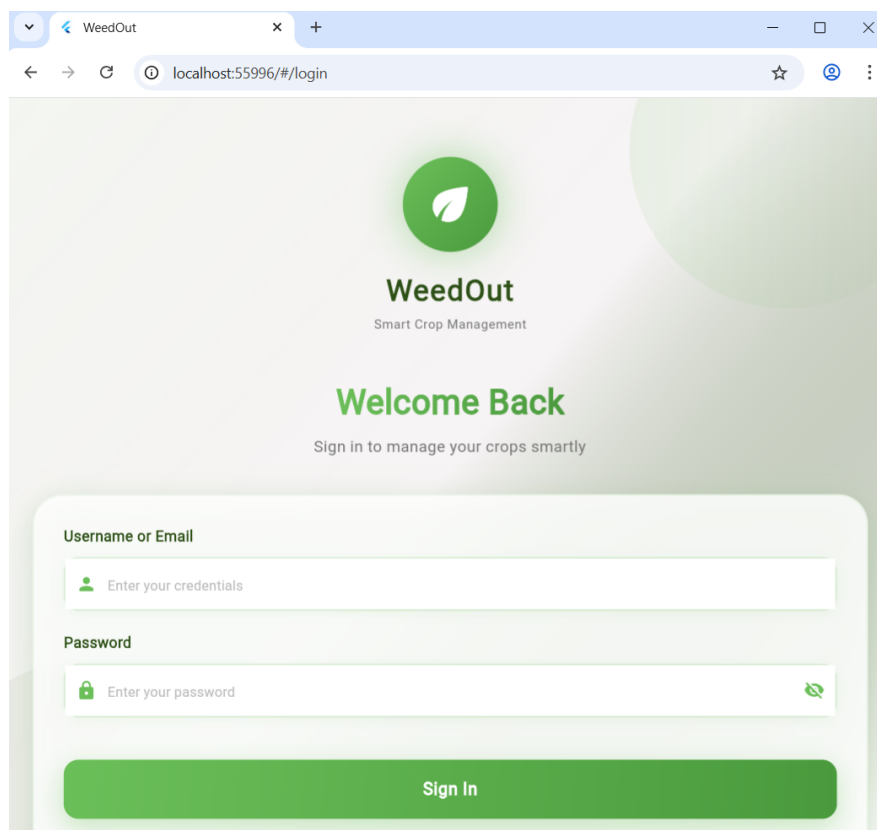
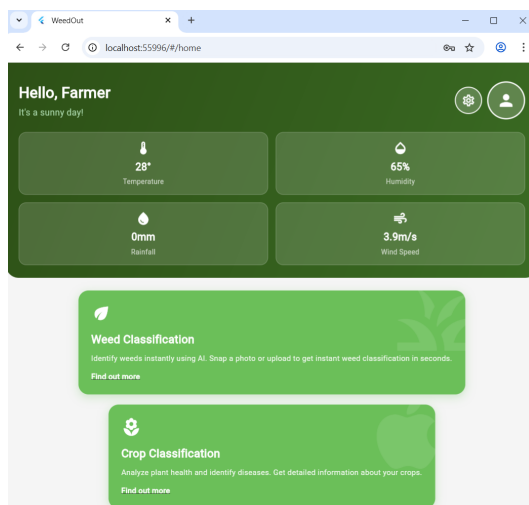
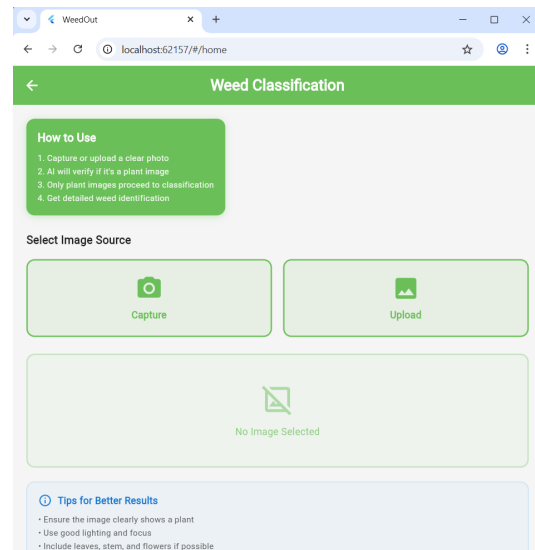


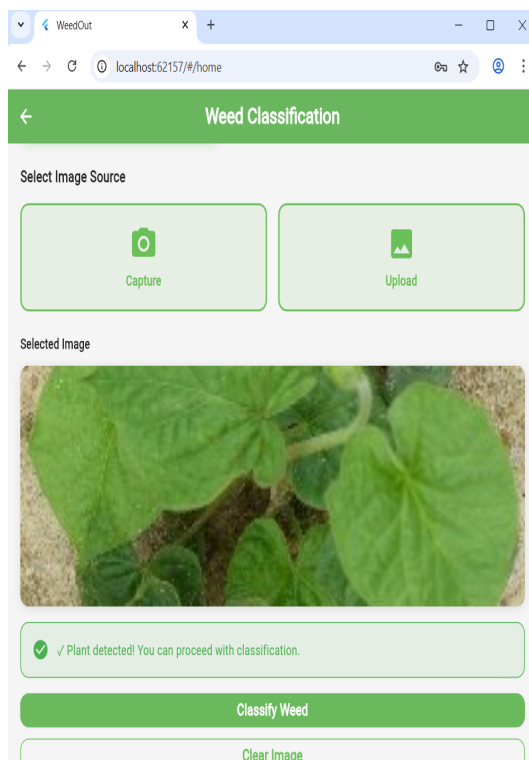
Figure 6.1: Login Page



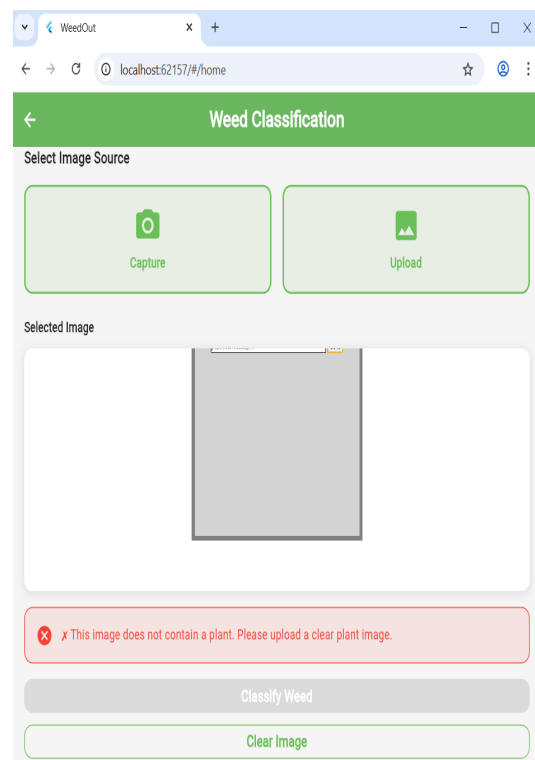
(a) Home Page



(b) Weed Classification Interface



(c) Contains a plant



(d) Does not contain a plant

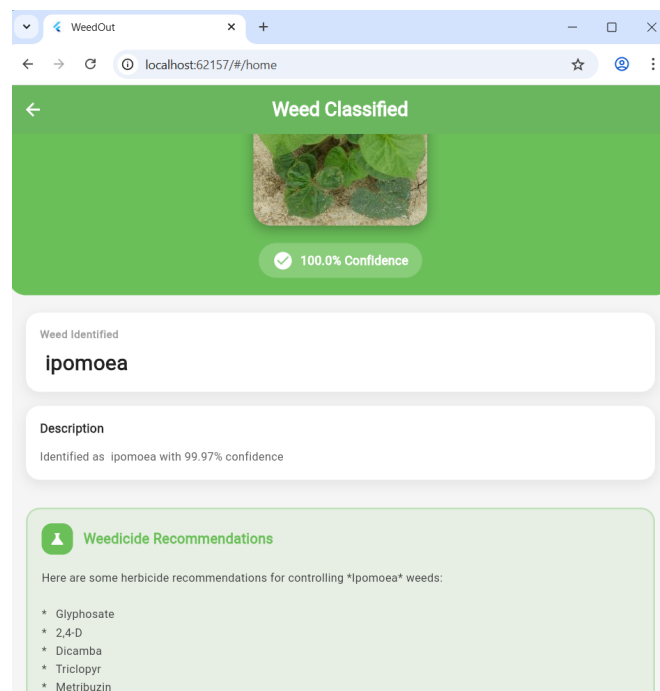
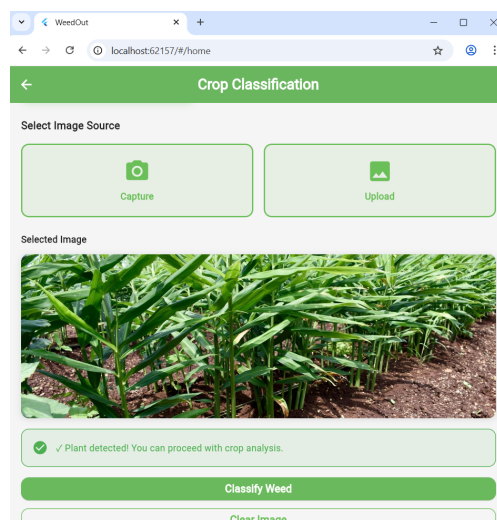
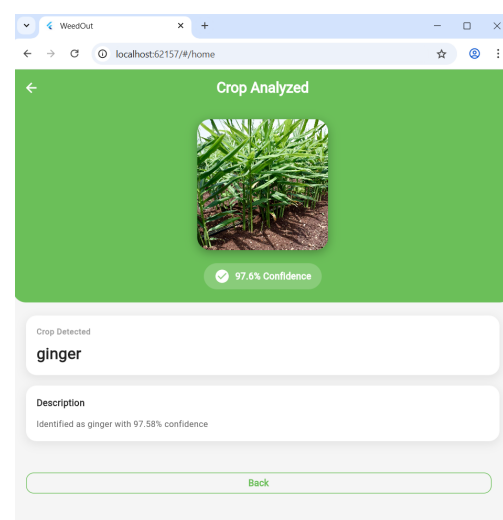


Figure 6.3: Ipomoea detected and weedicide is recommended



(a) Selected input image.



(b) Ginger crop detected

Figure 6.4: Crop classification interface

Step	Train Accuracy	Cross Entropy	Validation Accuracy	N
100	99.000%	0.180437	96.0%	100
200	98.000%	0.129043	99.0%	100
700	99.000%	0.064968	99.0%	100
800	99.000%	0.031480	98.0%	100

Figure 6.5: Performance Metrics

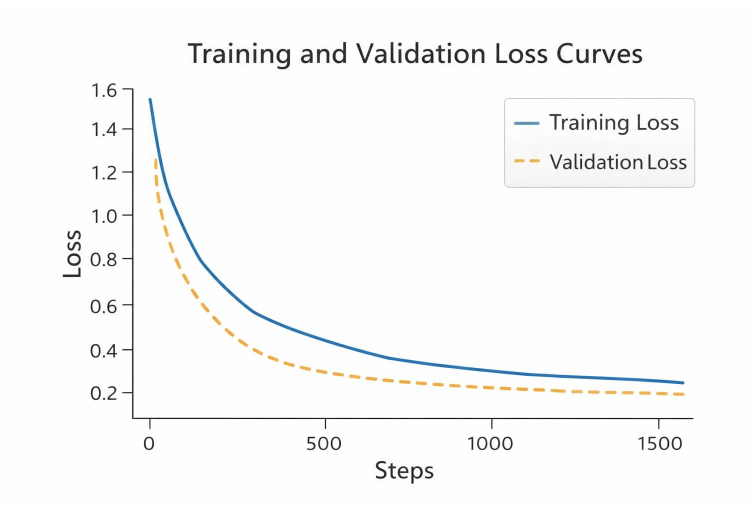


Figure 6.6: Loss curve.

Chapter 7

Conclusion

A new system has been developed to solve the main problems of Precision Agriculture through a deep learning (DL)-based model using convolutional neural networks (CNN) and combining it with appropriate image preprocessing techniques to improve classification accuracy for crops and weeds. This new model is trained with a diverse dataset that includes a variety of different crops (paddy, maize, tomato, ginger) as well as several types of weeds (including broadleaf, grass, and sedge) in order to produce a model that can be used successfully under a variety of conditions. The results of the experiments indicate that the CNN performs very well in terms of accurately distinguishing between crops and weeds with relatively high levels of reliability. Additionally, the inclusion of a user-friendly interface allows farmers who may have limited technical skills to use the farm, and also use of real-time image processing enables the model to be more useful in the field. The inclusion of a confidence score for each classification also allows the user to make informed decisions. By providing accurate weed identification and recommendations for weedicide applications, the system will also reduce the use of excess amounts of chemicals, decrease environmental impacts associated with chemical use, and promote sustainable practices in agriculture. This project demonstrates that the potential for AI (artificial intelligence) technologies to revolutionise and improve modern farming systems is enormous.

References

- [1] S. I. S. P. U. Nithin, S. M. A. Kareem, and G. V. Kailash, “Weed net: Deep learning informed convolutional neural network based weed detection in soybean crops,” pp. 1–8, 2023.
- [2] Y. Liu, S. Liu, J. Xu, X. Kong, L. Xie, K. Chen, Y. Liao, B. Fan, and K. Wang, “Forest pest identification based on a new dataset and convolutional neural network model with enhancement strategy,” *Computers and Electronics in Agriculture*, vol. 192, p. 106625, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0168169921006426>
- [3] A. S. M. M. Hasan, F. Sohel, D. Diepeveen, H. Laga, and M. G. Jones, “A survey of deep learning techniques for weed detection from images,” *Computers and Electronics in Agriculture*, vol. 184, p. 106067, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0168169921000855>
- [4] A. Ahmad, D. Saraswat, V. Aggarwal, A. Etienne, and B. Hancock, “Performance of deep learning models for classifying and detecting common weeds in corn and soybean production systems,” *Computers and Electronics in Agriculture*, vol. 184, p. 106081, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0168169921000995>
- [5] S. G. C, Y. Zhang, C. Koparan, M. R. Ahmed, K. Howatt, and X. Sun, “Weed and crop species classification using computer vision and deep learning technologies in greenhouse conditions,” *Journal of*

- Agriculture and Food Research*, vol. 9, p. 100325, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666154322000588>
- [6] A. Wang, W. Zhang, and X. Wei, “A review on weed detection using ground-based machine vision and image processing techniques,” *Computers and Electronics in Agriculture*, vol. 158, pp. 226–240, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0168169918317150>
- [7] N. Razfar, J. True, R. Bassiouny, V. Venkatesh, and R. Kashef, “Weed detection in soybean crops using custom lightweight deep learning models,” *Journal of Agriculture and Food Research*, vol. 8, p. 100308, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666154322000412>
- [8] X. Jin, J. Che, and Y. Chen, “Weed identification using deep learning and image processing in vegetable plantation,” *IEEE Access*, vol. 9, pp. 10 940–10 950, 2021.
- [9] V. Balafas, E. Karantoumanis, M. Louta, and N. Ploskas, “Machine learning and deep learning for plant disease classification and detection,” *IEEE Access*, vol. 11, pp. 114 352–114 377, 2023.
- [10] N. Nigar, H. Muhammad Faisal, M. Umer, O. Oki, and J. Manappattukunnel Lukose, “Improving plant disease classification with deep-learning-based prediction model using explainable artificial intelligence,” *IEEE Access*, vol. 12, pp. 100 005–100 014, 2024.

Publication

The research paper titled **”CNN-Based Weed Identification for Supporting Precision Weed Control”** was accepted for presentation at the International Conference on Emerging Technologies and Computing Innovations (ICETCI 2026), held on March 21–22, 2026. The conference is organized by MG Aricent Educational Foundation, Nashik, India. The paper will be published in the Springer STEAM-H Conference Proceedings, which is indexed in Scopus and other reputed databases.